

Air-cushion and Impact Force Coefficient with the Water Entry of a Flat Rigid Body

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ABSTRACT

The problem of water entry of a rigid body with low deadrise angle is numerically simulated by means of computational fluid dynamics. The Reynolds averaged Navier-Stokes equations are employed to simulate the water and the air. The profile of the static water surface would be deformed when the flat rigid body approaches to it. The air-cushion forms when the flat rigid body touches the water surface and it will expand and shrink alternatively. We mainly find that the average of the impact force coefficient is constant with fixed entry displacement, and the oscillating period of the impact force coefficient will decrease with the time going after the air-cushion is formed.

KEY WORDS: flat rigid body; water entry; air-cushion; numerical simulation; force coefficient

INTRODUCTION

The problems of structure water-entry slamming exist widely at the ocean engineering (Abrate, 2011; Faltinsen, 2015). When ships navigate under the atrocious sea conditions, the hull and other parts of ship will dash frequently with the water surface because of the significant pitching and heaving. Local structure failure or the entire hull vibration, even wreckage, will occur under the remarkable impacting forces (Panciroli, et al. 2013; Faltinsen, 2015).

Significant work on this water entry problem was firstly introduced by developing an analytical formula which allows estimation of the maximum pressure on seaplane floats during water landing (von Karman, 1929). Wagner (1936) modified the von Karman's solution by taking into account the effect of water splashing on the body. The first numerical complete solution to water entry was obtained by Dobrovol'skaya (1969).

Zhao & Faltinsen (1993) introduced a complement to Wagner's studies, with linear approximation of the free-surface boundary condition for the two-dimensional impact problem. Mei et al. (1999) proposed a purely analytical method of resolution for the global two-dimensional impact problem of arbitrary bodies. Moreover Yettou et al. (2007) took

into account the effect of velocity reduction of the solid body upon impact in order to determine impact pressure as well as the overall force acting on the body. While Lu et al. (2000) included the elastic deformation of the wedge in their analysis. Wu et al. (2004) carried out a numerical and experimental study of this problem. And a new iteration algorithm is proposed, in which the free surface boundary conditions were transformed to integral forms, and the developed jet was modelled through the shallow water equation. This method was extended by Xu et al. (2008) for the problem of oblique water entry of an asymmetric wedge. The problem was also solved by Semonov & lafrati (2006) using a kind of mapping method for the complex potential.

A high order boundary element method is employed by Wu (2012) to numerically simulate the water entry of a wedge at varying speed. A complex variable boundary element method is used by Gao et al. (2011) to simulate the water entry problem and the simulation results are compared with the similarity solution. Numerical simulations of water entry of a single wedge and twin wedges with three degrees of freedom are carried out by Xu et al. (2009, 2011). The similarity method was developed to solve the three-dimensional oblique water entry (Wu & Sun, 2014) and the collision of two liquid wedges (Semenov, et al. 2013) or liquids with different densities (Semenov, et al. 2015).

The air pocket in flat impact of box type model was tested by Oh et al. (2009, 2010), and the side view and bottom view of the flat impact were captured with high speed camera. When a flat rigid body impacts the static water surface, or enters water with a constant velocity or freely falling down, air-cushion maybe be generated, so the air flow, even the compressibility of the air must be taken into account in these similar models. A fine numerical simulation will be requested for the small space scale of the air-cushion and the small-time scale of its generation and diminish.

NUMERICAL METHOD

Problem Description

The problem in this paper can be described as follows: a two-dimensional flat rigid body, which length is 0.28m and width is 0.07m,

falls down from the height 0.13m above the static water surface with constant velocity, and the water depth is 0.3m. The elastic of the body is ignored and the flat body can be regarded as rigid body and the entry water velocity is dominated as a constant. The problem description is shown as Fig. 1.

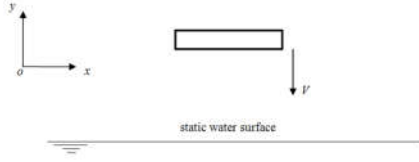


Fig. 1 Sketch of the problem

Governing Equations

The flow of fluids, including air and water, can be described by the equation of mass conservation, or continuity equation as

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0, \quad (1)$$

and the momentum conservation equation as

$$\frac{\partial \rho \vec{v}}{\partial t} + \nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla p + \nabla \cdot \vec{\tau} + \rho \vec{g} \quad (2)$$

where t is time, ρ is the fluid density, $\vec{v} = (u, v, w)$ is the velocity vector, p is the static pressure, $\vec{\tau}$ is the stress tensor and $\rho \vec{g}$ is the gravitational body force.

In this paper, two-dimension, unsteady, double precision, pressure-based solver, RNG $k-\epsilon$ turbulence models are selected. PISO scheme for pressure-velocity coupling calculation and body force weighted scheme are active to solve and the geo-reconstruct scheme is enabled to capture the free surface. The solutions are considered to be converged when the residuals were less than 10^{-3} , and time step size is 10^{-6} s. The simulations are performed at a PC with eight 3.4 GHz processors and 16 G bytes RAM.

For the flat rigid body falls down with a constant velocity, the dynamic grid method is employed and the whole grid is moved at the same velocity but the flow field is maintained non-motive with the flat body. The costumed time dependent boundary and volume of fraction conditions should be imposed on the pressure-in and pressure-out boundaries.

NUMERICAL RESULTS

Flat Rigid Body Impacting on the Water Surface

We focus on the water-entry of flat rigid body, which differs distinctly from the one of wedge with large deadrise angle. In the process of water entry, there are several stages including generation of ripples around the edge of the flat body, generation of air-cushion (air pocket) and diminish of it between the flat body and the water surface.

The comparisons of the profile of water surface at different stages between numerical simulation and experiment with the same parameters are shown in Fig.2~3.

We can see from Fig.2 that the water ripples will generate around the edge when the flat body approaches to water surface and there is a

small gap between the water surface and the flat rigid body. Fig.3 shows that the deformed water surface induced by the nonuniform distribution of pressure and the bottom of flat rigid body will form air-cushion.



(a) Experiment



(b) Numerical simulation
Fig. 2 Generation of ripples



(a) Experiment



(b) Numerical simulation
Fig. 3 Generation of air-cushion

Because of the compressibility of the air and the high pressure in the air-cushion, the configuration of the air-cushion would change with time going. This means that the air-cushion will expand and shrink at different moment with the changing of the air pressure in it. Fig.4 presents the expansion and shrink of air-cushion at the first time and Fig.5 at the second time.



(a) Expansion (t=0.0680s)



(b) Shrink (t=0.0710s)

Fig. 4 Expansion and shrink at the first period



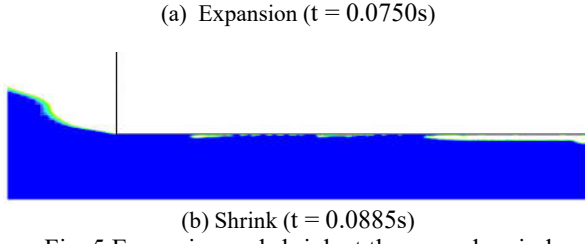


Fig. 5 Expansion and shrink at the second period

In the experiment, the air-cushion would diminish in a short time when the flat body pierces into the water surface. But in our numerical simulation the air-cushion doesn't diminish within a short time. The reasons for this different would be: 1) the air dissolution isn't considered and no model of dissolution is included in this paper, and 2) the deformation of flat body and the three-dimensional influence aren't taken into account in the numerical simulation, but in the experiment the flat body would be deformed because of the high pressure and its nonuniform distribution and the air would overflow at another dimension.

Velocity Distribution

In order to analyze the air velocity distribution at the bottom of the flat body, a simple mathematical model is set up. Obviously, the air at the left of symmetry will be pushed out towards left. Supposed that the air velocity distributes uniform along the y axis, the water surface keeps static and the air is incompressible, like as Fig. 6.

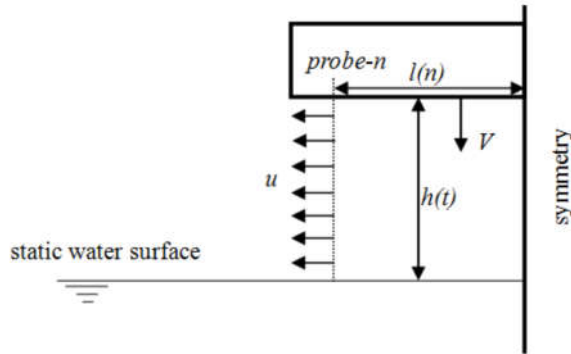
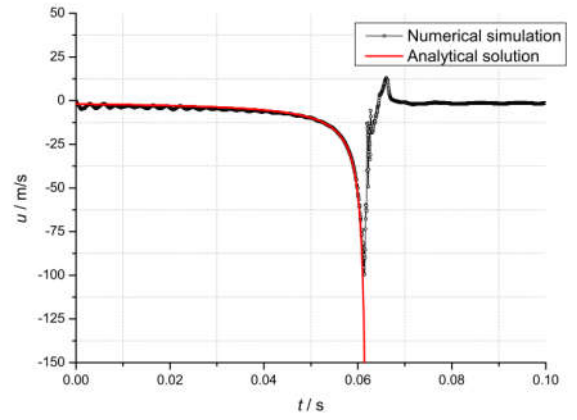


Fig. 6 Simple model for water entry of flat body

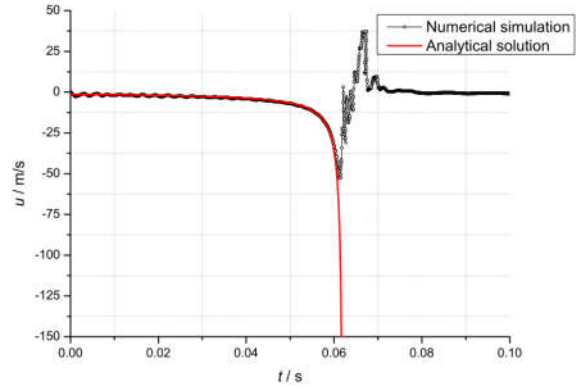
From the above assumption, we can get that the velocity at the probe n , which is $l(n)$ m horizontally from the symmetry and $h(t)$ m vertically from the bottom of flat body, can be written as

$$u(n, t) = -\frac{V \cdot l(n)}{h(t)}. \quad (3)$$

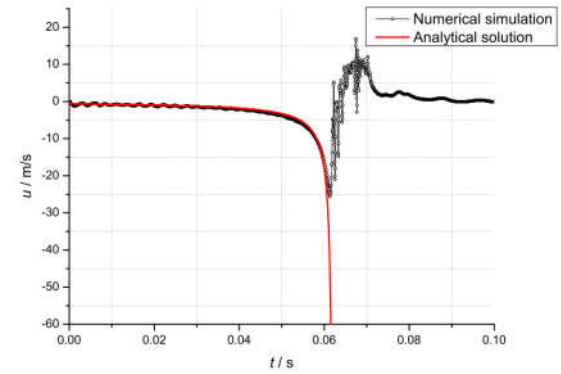
In this paper, three probes are located at the bottom of the rigid body and $l(n)$, here $n = 1, 2, 3$, are equal to 0.12m, 0.08m and 0.04m, respectively. Fig.7 (a)~(c) are the velocity of x direction for probes 1~3. These figures show that the x velocity of numerical simulations is coincided with those of analyses before the flat rigid body contacts with the free surface when the simulation time is equal to 0.062s. But when the flat body touches the free surface and the air-cushion has generated, the x velocity drops sharply and fluctuates irregularly.



(a) Probe 1



(b) Probe 2



(c) Probe 3

Fig. 7 Time history of x velocity at various probes

From Fig.7(a) we can see that the x velocity is close to 100m/s, which is very high speed compared to the drop speed of the rigid body. The relative higher velocity at probe 1 compared to the other two probes would lead to lower air pressure and give rise to the nonuniform distribution of the pressure at the bottom of the rigid body. Thus, the ripple will generate and then the air-cushion do, as shown in Fig.2 and Fig.3 including experimental snapshots and numerical simulations.

Impact force coefficient

The time history of calculational pressure in probe 1 is compared with

that of experimental one, as shown in Fig. 8. The general tendencies of change are similar. But the maximum pressure in experiment is less than the one in calculation and the ratio between them is about 0.36. Moreover, the amplitude of fluctuation in experiment is larger than the calculational one. The reasons for these different are that the constant drop velocity of flat body would decrease in the experiment when it contacts the free surface because of the high pressure on the bottom of the rigid body, and it is not a strict two-dimensional flat rigid body, which would take the elasticity of the structure into account. Another reason for experiment not to capture the high peak of the pressure may be that the experiment hasn't enough time resolution to detect it.

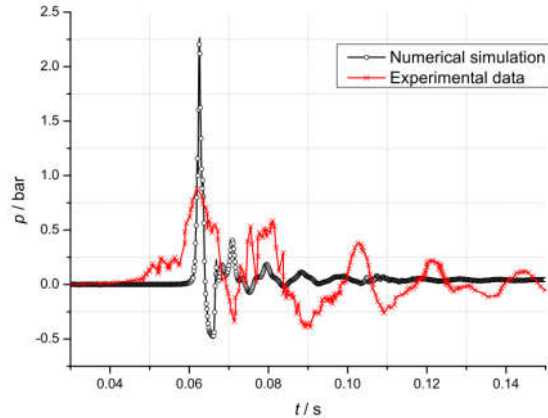


Fig. 8 Numerical result with experimental data

The time history of the impact force coefficients with various water entry velocities is shown in Fig. 9. It can be seen that the impact force coefficients present dominating time domain characteristics. Larger the impact velocities are, shorter the impact last time is.

The impact force coefficient is integrated over time and the value multiples the square of velocity is calculated and this parameter is plotted with velocity in Fig. 10. It shows that the relationship between this parameter and impact velocity is perfect linearity. Undouble, the relationship between the average impact force coefficient and the velocity would be also linear, as shown in Fig. 11.

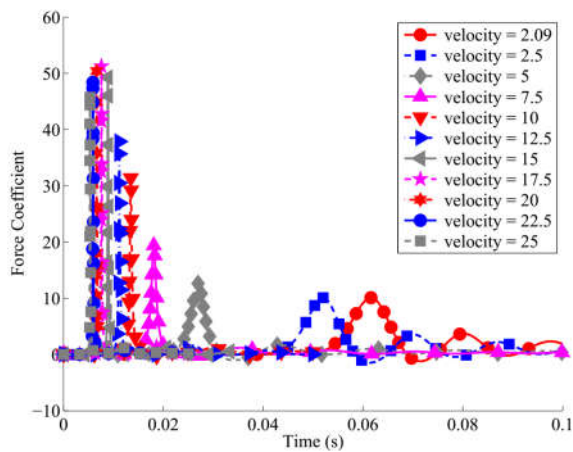


Fig. 9 Time varieties of the impact force coefficients with various velocities

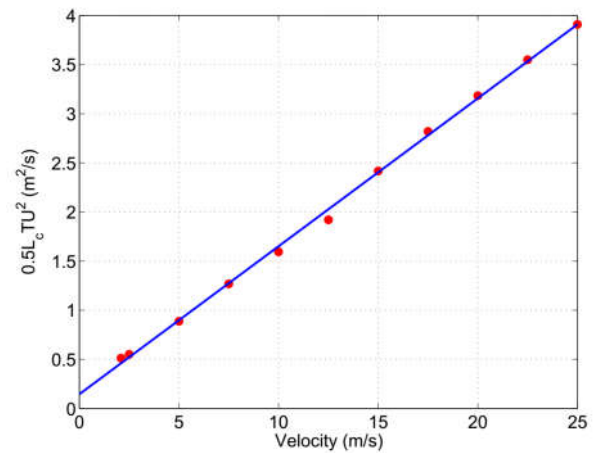


Fig. 10 Relationship between impact velocity and integration of force coefficient

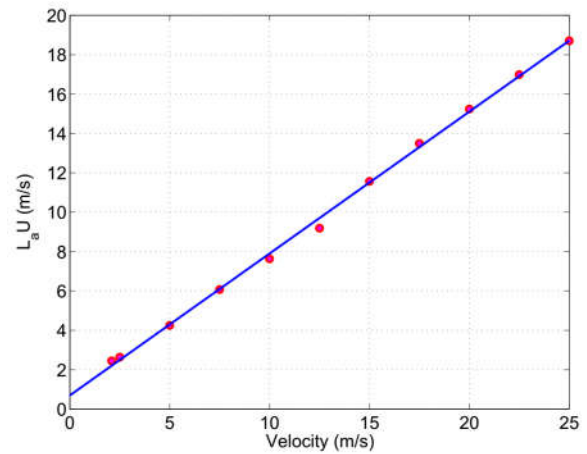


Fig. 11 Relationship between impact velocity and average of force coefficient

The relationship between the peak value of the impact coefficient and the entry velocity is shown in Fig. 12 and it shows that the peak value will increase with that of the velocity below 22.5m/s and it will decrease with that over 22.5 m/s because of the effect of the air-cushion.

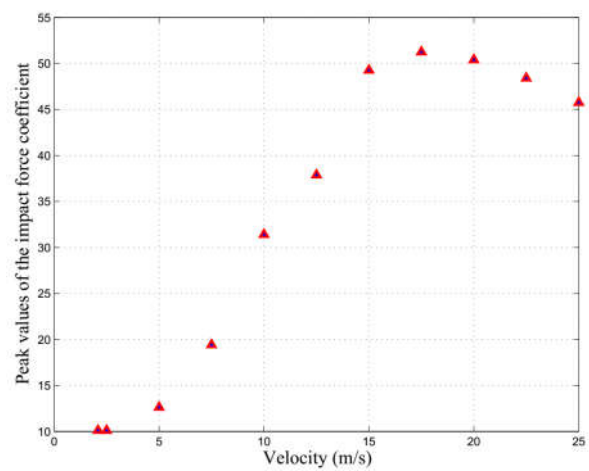


Fig. 12 Relationship between peak value and velocity

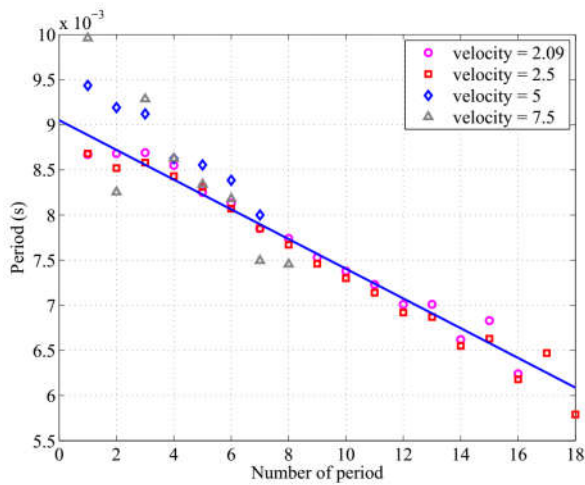


Fig. 13 Period of the oscillating force coefficients

The frequency or the time period, which is dominated by von Karman vortex street, of the lift force or drag force is a constant with certain inflow in the case of cylinder. But in these cases of water entry with air-cushion, the time period is no longer a constant, as shown in Fig.13. The time period will decrease with the number of period and the relationship between them is linear.

CONCLUSIONS

A fine numerical simulation to the water-entry of low deadrise angle rigid body, especially the flat rigid body is carried out by means of computational fluid dynamics, including the turbulent model, the compressibility of the air, etc.

The numerical simulations fit well with the analytical and experimental results when the deadrise angle is large. But the maximum of pressure coefficient exists significant difference when the deadrise angle is low because of the generation of the air-cushion. Prominent air-cushion is generated when the flat body impacting the water surface and the shape of the air-cushion will experience the process of expand and shrink. The pressure and x direction velocity of air will reach its peaks just before the generation of air-cushion and then them decrease dramatically.

Analyzing on the impact force coefficient shows that the average of it would be a constant in spite of the variety of the impact or entry velocity. And the impact force coefficient will be oscillating with different period after the flat body impacts the water surface and the period decreases linearly with time going.

In this paper, the acceleration and the elasticity of the rigid body are not considered. The characteristics of the cavitation, when the rigid body immerses into the water deeply, is also our aims to investigate using computational dynamics tool in the future.

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